

GENERATIVE AI FUNDAMENTALS

Introduction

Generative Artificial Intelligence (GenAI) is a type of AI that can create new content such as text, images, audio, videos, and code based on user input.

Applications of GenAI

- Content generation
- Image creation
- Chatbots and virtual assistants
- Code generation
- Language translation

Popular GenAI Tools

- ChatGPT
- Google Gemini
- GitHub Copilot
- Claude
- DALL-E

Prompt Engineering

Prompt Engineering is the process of designing effective prompts to obtain accurate responses from AI systems.

Types of Prompting

Zero-Shot Prompting

Providing a task without examples.

Few-Shot Prompting

Providing a few examples before the task.

Chain-of-Thought Prompting

Breaking a problem into step-by-step reasoning.

Benefits of GenAI

- Increased productivity
- Faster content creation
- Automation of repetitive tasks
- Improved creativity

Challenges of GenAI

- Privacy concerns
- Security risks
- Bias in AI models
- Incorrect information (Hallucinations)

Conclusion

Generative AI is transforming industries by enabling intelligent content creation and automation, making it one of the most important technologies of the future.